Transformations and Inverse Relations

Concept

Transformations move or scale a graph. Inverses swap its inputs and outputs.

Point mapping

For $y = a\,f(b(x-h)) + k$: $(u,v) \to (u/b+h,\ av+k)$, $a,b \neq 0$.

Inside change

Factor first: $f(2x - 6) = f(2(x - 3))$; shift right 3.

Inverse

Swap x,y and solve; use the horizontal-line test or restrict domain.

Domain / range

Transform allowed coordinates; inverse swaps domain and range.

Example 1

Map $(2,3)$ under $y = -2f(3(x - 4)) + 1$.

Solution

1. New $x = 2/3 + 4 = 14/3$.
2. New $y = -2(3) + 1 = -5$.
3. Mapped point: $(14/3,-5)$.

Transformations and Inverse Relations

Example 2

Find the inverse of $f(x) = (x - 2)^2 + 1$, restricted to $x \geq 2$.

Solution

1. Swap variables: $x = (y - 2)^2 + 1$.
2. Since $y \geq 2$, take the positive root: $y = 2 + \sqrt{(x - 1)}$.
3. $f^{-1}(x) = 2 + \sqrt{(x - 1)}$; domain $x \geq 1$, range $y \geq 2$.

Example 3

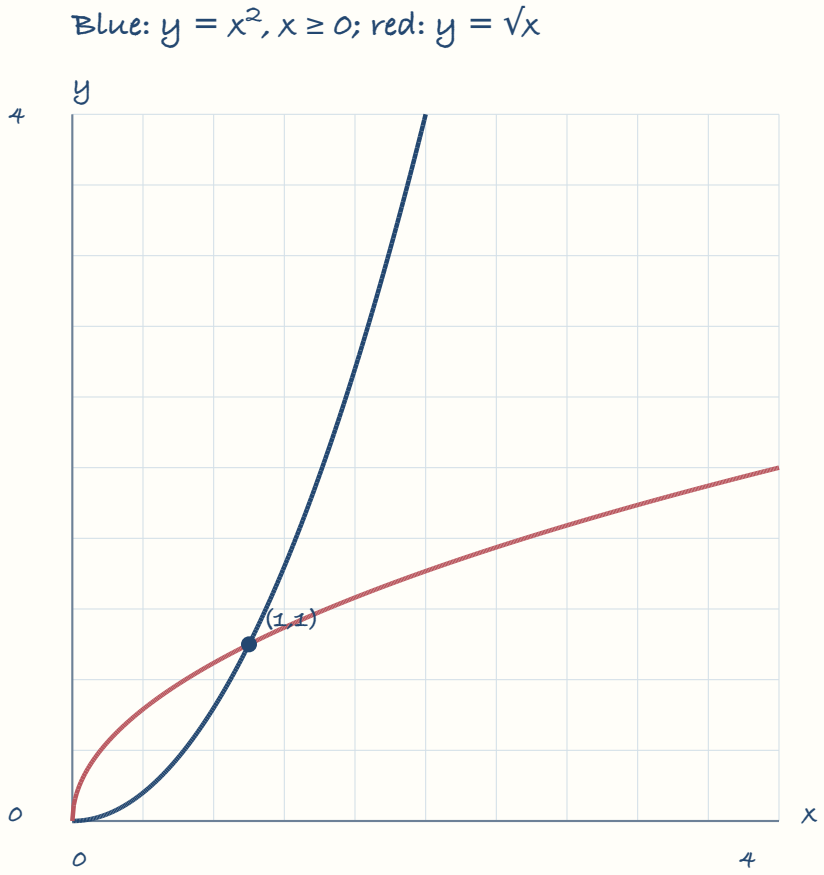
Describe $y = |x - 2| - 3$.

Solution

1. Move $|x|$ right 2 and down 3: vertex $(2,-3)$.
2. Domain all real x; range $y \geq -3$.
3. Piecewise: $-x - 1$ for $x < 2$; $x - 5$ for $x \geq 2$.

Transformations and Inverse Relations

Visual notes | read the graph



Equal axis scales show reflection across $y = x$. The restricted parabola and square-root graph are inverse functions.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Transformations and Inverse Relations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Map $(4,-1)$ under $y = 3f(2(x + 1)) - 2$.

2. State domain and range of $y = 2\sqrt{(x + 3)} - 4$.

3. Find the inverse of $f(x) = (x - 5)/2$.

4. Why is $y = x^2$ not one-to-one on the whole real line?

Transformations and Inverse Relations

Practice answers | check after trying

1. $(4/2 - 1, 3(-1) - 2) = (1,-5)$.

2. Domain $[-3,\infty)$, range $[-4,\infty)$.

3. $f^{-1}(x) = 2x + 5$.

4. Opposite inputs such as 2 and -2 give the same output 4, so the inverse relation would have two outputs for input 4.

Polynomial Functions and Equations

Concept

Zeros, multiplicities and the leading term explain the main shape of a polynomial graph.

End behaviour

Degree parity and leading-coefficient sign control the two ends.

Multiplicity

Odd zero: crosses; even zero: touches the x-axis.

Factor / remainder

Dividing by $x - a$ leaves remainder $P(a)$; $P(a) = 0$ means a factor.

Division

Include zero coefficients for missing powers in synthetic division.

Example 1

Factor $P(x) = x^3 - 4x^2 + x + 6$ and solve $P(x) = 0$.

Solution

1. Test $x = 2$: $P(2) = 0$, so $x - 2$ is a factor.
2. Synthetic division: coefficients $1, -4, 1, 6$ with 2 give $1, -2, -3$ and remainder 0.
3. Quotient $x^2 - 2x - 3 = (x - 3)(x + 1)$.
4. $P(x) = (x - 2)(x - 3)(x + 1)$; roots $-1, 2, 3$.

Polynomial Functions and Equations

Example 2

Describe $y = -(x + 1)^2(x - 2)$.

Solution

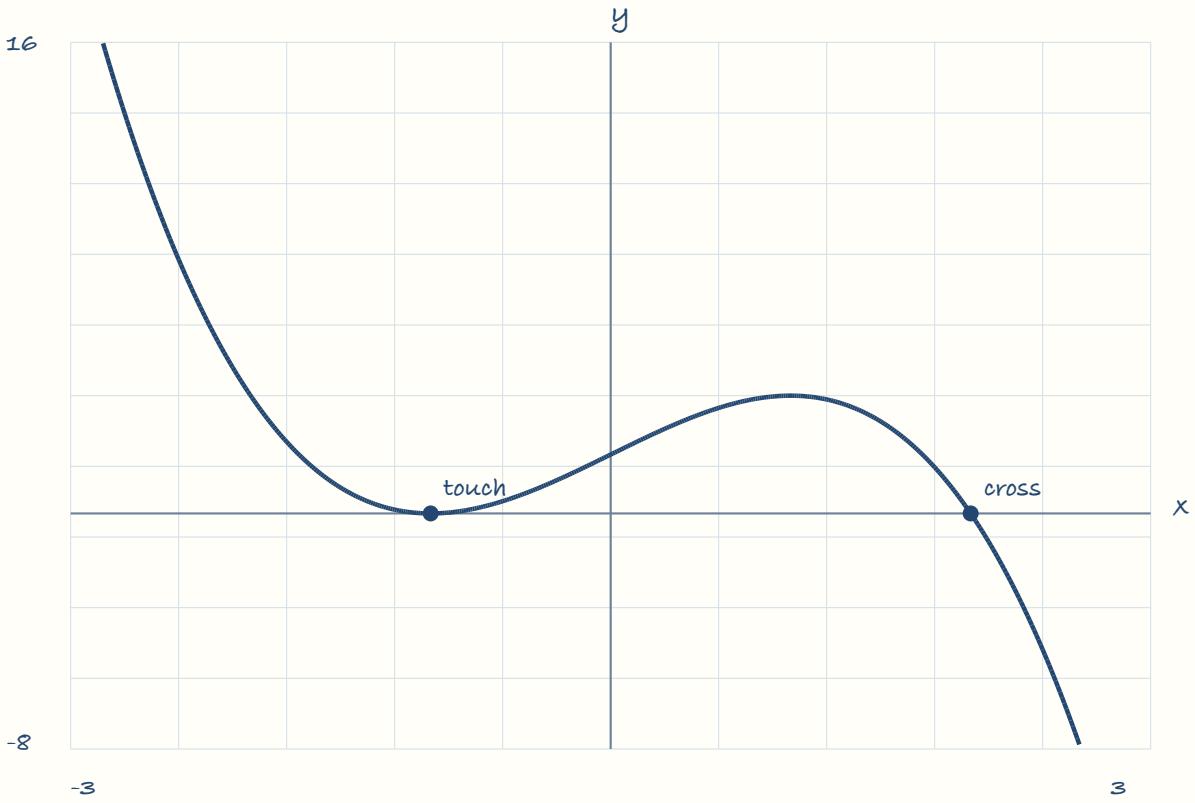
1. Degree 3, negative leading coefficient: left end up, right end down.

2. Touch at $x = -1$; cross at $x = 2$. y-intercept $= 2$.

Polynomial Functions and Equations

Visual notes | read the graph

$$y = -(x + 1)^2(x - 2)$$



The double zero touches. The simple zero crosses. The leading negative cubic term controls the ends.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Polynomial Functions and Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the remainder of $x^3 + 2x^2 - 5$ at division by $x - 1$.

2. Describe ends and zero behaviour of $y = 2(x - 1)^2(x + 3)^2$.

3. Factor $x^3 - 3x^2 - 4x + 12$.

4. Solve $x^3 = 4x$.

Polynomial Functions and Equations

Practice answers | check after trying

1. $P(1) = 1 + 2 - 5 = -2$.

2. Degree 4, leading coefficient positive: both ends up. It touches at -3 and 1.

3. Grouping gives $x^2(x - 3) - 4(x - 3) = (x - 3)(x - 2)(x + 2)$.

4. $x(x^2 - 4) = x(x - 2)(x + 2) = 0$, so $-2, 0, 2$. Dividing by x without checking would lose the zero solution.

Rational Functions and Discontinuities

Concept

Rational graphs can have holes and asymptotes. Start with the original denominator.

Domain

Exclude every original denominator zero before cancelling factors.

Hole / vertical asymptote

A fully cancelled factor gives a hole; a remaining denominator zero gives an asymptote.

Horizontal asymptote

Lower numerator degree: $y=0$; equal degrees: leading-coefficient ratio.

Slant asymptote

If numerator degree is one larger, divide to find a linear quotient.

Example 1

Find the key features of $(x^2 - 1)/(x^2 - x - 2)$.

Solution

1. Factor and cancel $x+1$: $(x-1)/(x-2)$, with $x \neq -1,2$.
2. Hole at $(-1,2/3)$; vertical asymptote $x=2$; horizontal asymptote $y=1$.
3. Intercepts: $(1,0)$ and $(0,1/2)$.

Rational Functions and Discontinuities

Example 2

Find the asymptotes of $y = (2x + 3)/(x - 1)$.

Solution

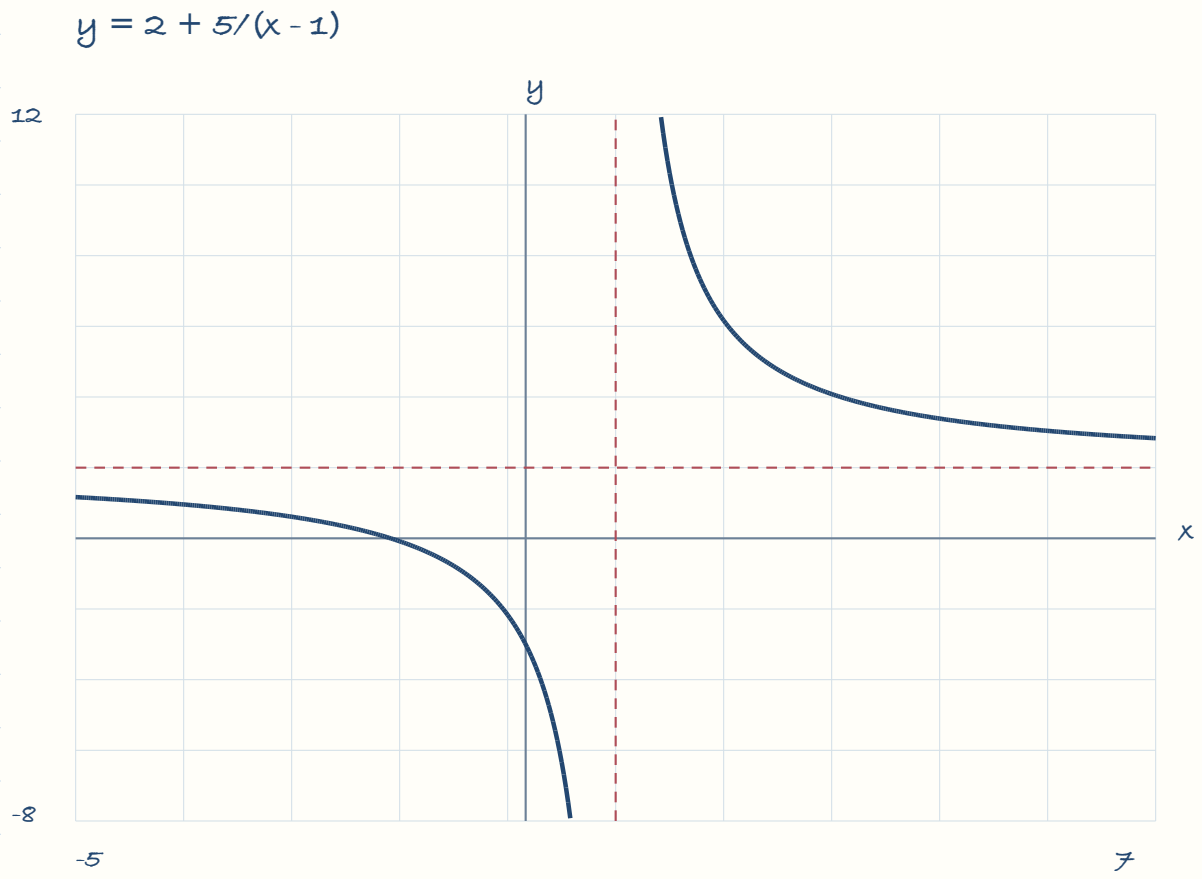
1. Rewrite: $y = 2 + 5/(x - 1)$.

2. Vertical asymptote $x = 1$; horizontal asymptote $y = 2$.

3. Domain $x \neq 1$; range $y \neq 2$.

Rational Functions and Discontinuities

Visual notes | read the graph



Dashed asymptotes: $x = 1$ and $y = 2$. Neither value is attained as the corresponding input or output.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Rational Functions and Discontinuities

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Describe the graph features of $(x^2 - 4)/(x - 2)$.

2. Find vertical and horizontal asymptotes of $(3x + 1)/(2x - 4)$.

3. Find intercepts of $(x - 3)/(x + 2)$.

4. Can $y = 1/(x^2 + 1)$ have a vertical asymptote over the reals?

Rational Functions and Discontinuities

Practice answers | check after trying

1. Line $y = x + 2$ with hole $(2,4)$; domain $x \neq 2$. No vertical asymptote at 2 because the denominator factor cancels completely.

2. $x = 2$; $y = 3/2$.

3. $(3,0)$ and $(0,-3/2)$.

4. No. $x^2 + 1$ is positive for every real x, so the denominator never vanishes.

Exponential Functions and Models

Concept

Exponential models multiply by a constant factor over equal time intervals.

Parent graph

$y=b^x$, $b>0$, $b\neq1$: domain all real x, range $y>0$, asymptote $y=0$.

Growth / decay

Multipliers $1+r$ / $1-r$; match the rate's time period.

Solve

Rewrite with the same base, or take logs after isolating the exponential.

Half-life

$A(t) = A_0(1/2)^{t/H}$, with half-life H.

Example 1

Solve $4^{x-1} = 8$.

Solution

1. Write powers of 2: $2^{2x-2} = 2^3$.

2. Equate exponents: $2x - 2 = 3 \rightarrow x = 5/2$.

Exponential Functions and Models

Example 2

A 160 g sample has a half-life of 3 hours. Find its mass after 9 hours.

Solution

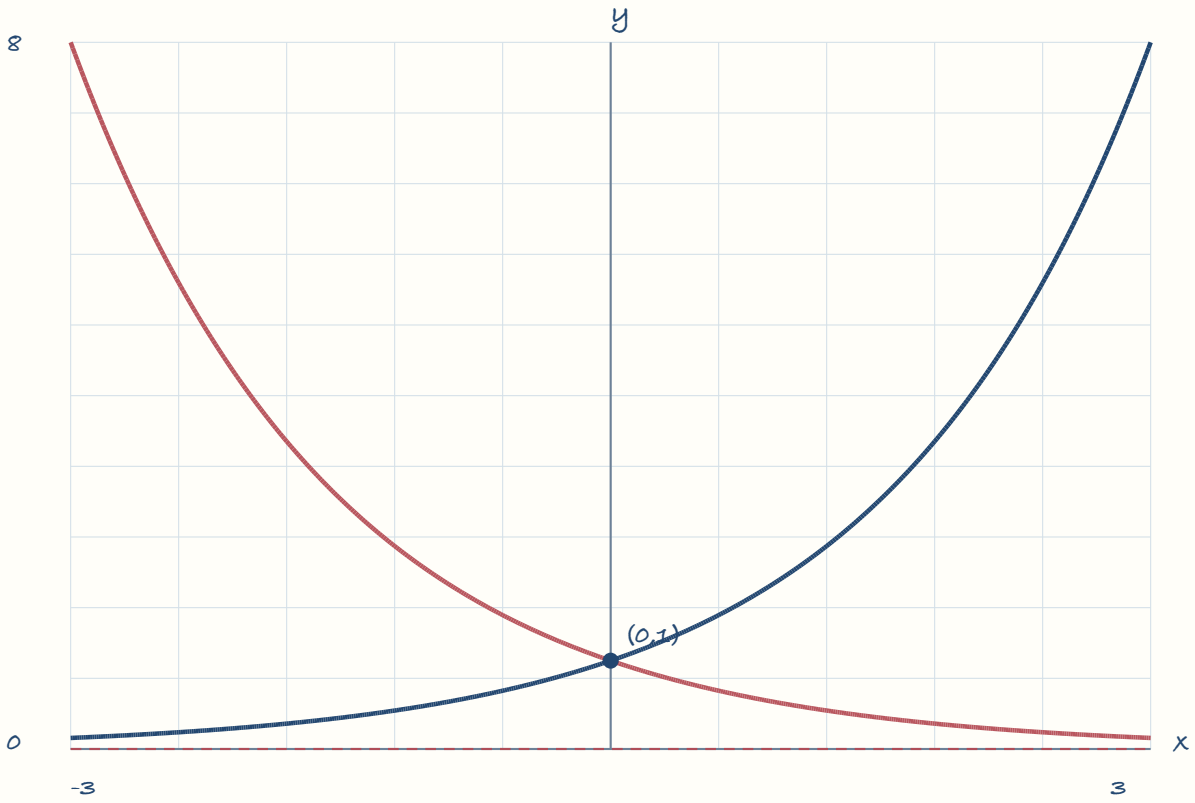
1. Number of half-lives $= 9/3 = 3$.

2. Mass $= 160(1/2)^3 = 20$ g.

Exponential Functions and Models

Visual notes | read the graph

Blue: $y = 2^x$; red: $y = (1/2)^x$



Both stay positive. One grows and the other decays. The horizontal asymptote is $y = 0$.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Exponential Functions and Models

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $9^x = 27$.

2. Write a model for \$500 losing 12% of value each year.

3. Find the mass after 8 h if 96 g has a half-life of 4 h.

4. State domain, range and asymptote of $y = 5 \times 2^x - 3$.

Exponential Functions and Models

Practice answers | check after trying

1. $3^{2x} = 3^3$, so $x = 3/2$.

2. $V(t) = 500(0.88)^t$, $t \geq 0$ in years.

3. Two half-lives: $96/4 = 24$ g.

4. Domain all real x, range $y > -3$, asymptote $y = -3$.

Logarithms and Logarithmic Equations

Concept

A logarithm answers: what exponent produces this number?

Definition

$\log_b x = y$ means $b^y = x$; $b > 0$, $b \neq 1$, $x > 0$.

Log laws

Product → sum; quotient → difference; power → multiplier, for positive arguments.

Change of base

$\log_b x = \ln x / \ln b$. log uses base 10; ln uses base e.

Restrictions

Every original log argument must be positive; check candidate roots.

Example 1

Solve $\log(x - 1) + \log(x + 2) = 1$.

Solution

1. Domain: $x > 1$. Combine: $\log[(x - 1)(x + 2)] = 1$.
2. $(x - 1)(x + 2) = 10 \rightarrow (x + 4)(x - 3) = 0$.
3. Candidates $-4, 3$; only $x = 3$ fits the domain.

Logarithms and Logarithmic Equations

Example 2

Solve $5 \times 2^x = 30$.

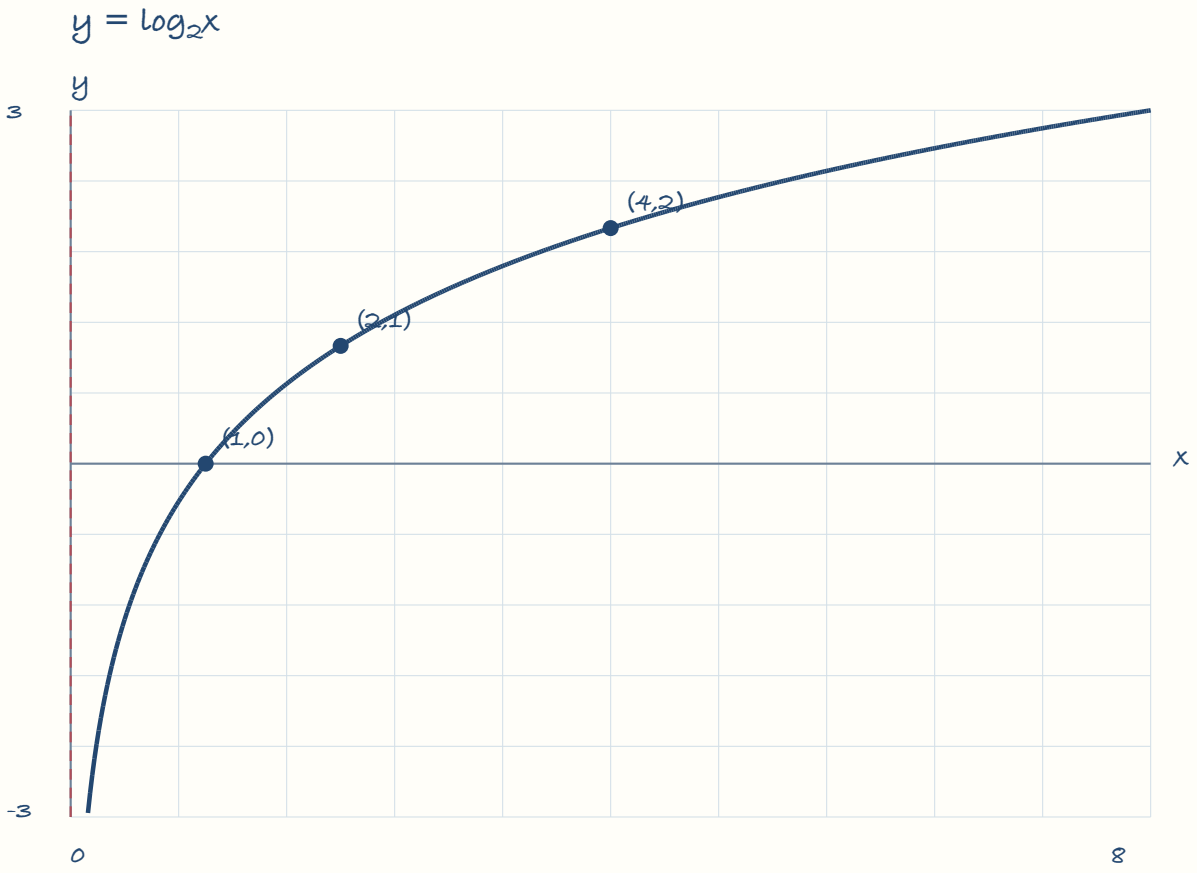
Solution

1. $2^x = 6 \rightarrow x \ln 2 = \ln 6$.

2. $x = \ln 6 / \ln 2 \approx 2.585$.

Logarithms and Logarithmic Equations

Visual notes | read the graph



The logarithm is undefined for $x \le 0$. Doubling the input adds 1 to the output.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Logarithms and Logarithmic Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Evaluate $\log_4 64$.

2. Expand $\ln(3x^2/y)$ for $x, y > 0$.

3. Solve $\ln(x - 2) = 0$.

4. Solve $3^x = 20$ exactly using logs.

5. Solve $\log_2 x + \log_2(x - 2) = 3$.

Logarithms and Logarithmic Equations

Practice answers | check after trying

1. 3.

2. $\ln 3 + 2\ln x - \ln y$.

3. $x - 2 = e^0 = 1$, so $x = 3$.

4. $x = \ln 20 / \ln 3 \approx 2.727$.

5. $x > 2$. $x(x - 2) = 8$ gives $(x - 4)(x + 2) = 0$. Only $x = 4$ is valid.

Geometric Sequences and Series

Concept

A geometric sequence has a constant ratio. A series adds its terms.

Term rule

$a_n = a_1 r^{n-1}$; a negative ratio alternates signs.

Finite sum

$S_n = a_1(1-r^n)/(1-r)$, $r \neq 1$; when $r=1$, $S_n = na_1$.

Infinite sum

$S_\infty = a_1/(1-r)$, only when $|r|<1$ for non-zero a_1.

Sigma notation

Count both index endpoints: $k=0$ to 3 contains four terms.

Example 1

Find term 7 and the sum of 7 terms of 5,15,45,... .

Solution

1. $r = 3$; $a_7 = 5 \times 3^6 = 3645$.

2. $S_7 = 5(3^7 - 1)/(3 - 1) = 5465$.

Geometric Sequences and Series

Example 2

Find the infinite sum $12 + 6 + 3 + \ldots$.

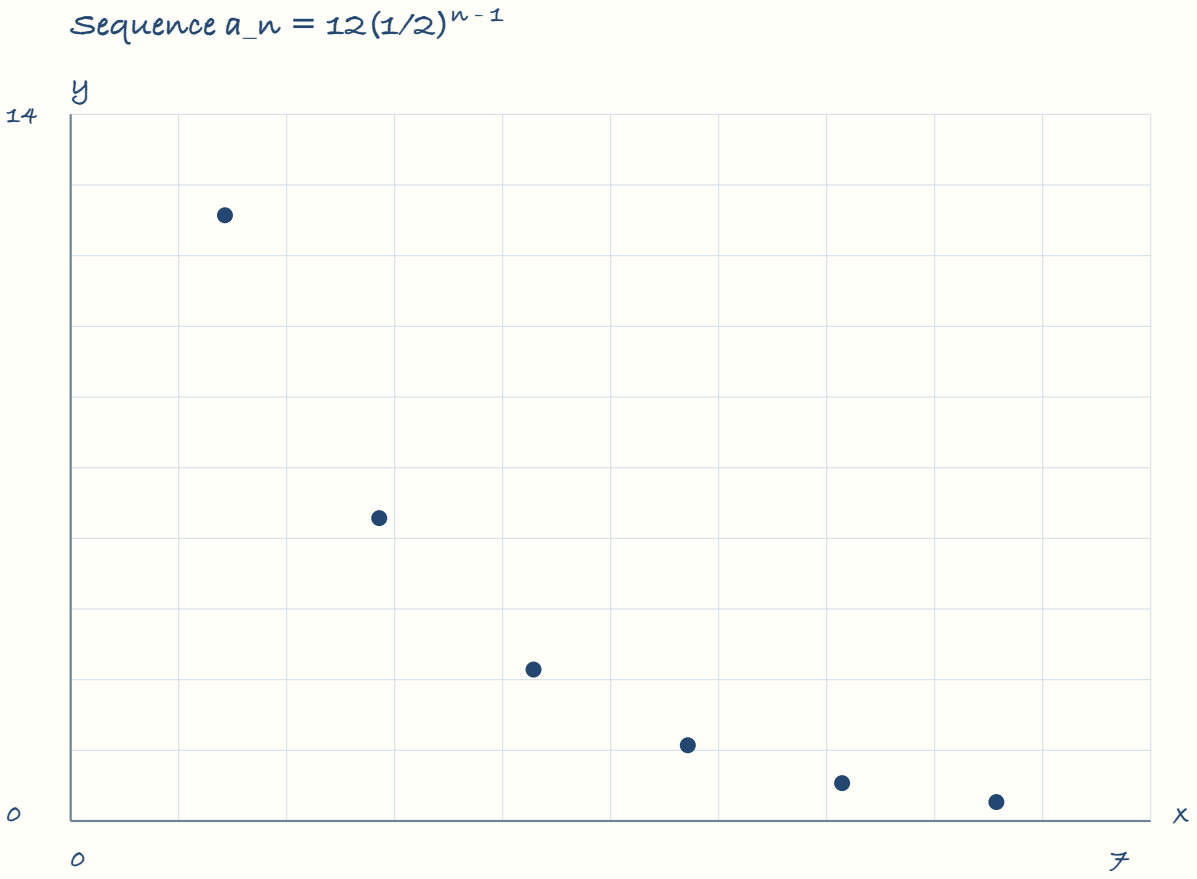
Solution

1. $r = 1/2$; $|r| < 1$, so the series converges.

2. $S_\infty = 12/(1 - 1/2) = 24$.

Geometric Sequences and Series

Visual notes | read the graph



Term magnitudes approach zero. The sum of the infinite geometric series approaches 24.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Geometric Sequences and Series

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find term 6 of 64,32,16,… .

2. Sum the first five terms of 2,6,18,… .

3. Find the infinite sum $9 - 3 + 1 - 1/3 + \ldots$.

4. Does $4 + 6 + 9 + \ldots$ converge?

5. Expand $\sum$ from $k = 0$ to 3 of $2(5^k)$.

Geometric Sequences and Series

Practice answers | check after trying

1. $64(1/2)^5 = 2$.

2. $2(3^5 - 1)/(3 - 1) = 242$.

3. $r = -1/3$, so $S\infty = 9/(1 + 1/3) = 27/4$.

4. No: $r = 1.5$ and terms do not approach zero.

5. $2 + 10 + 50 + 250 = 312$. There are four terms because both endpoints are included.

Radians and the Unit Circle

Concept

Radians measure angles using arc length divided by radius.

Convert

180° = π radians; degrees × π/180 = radians.

Circle measures

Arc $s=r\theta$; sector area $A=r^2\theta/2$. These formulas need radians.

Unit circle

$(\cos\theta, \sin\theta)$; use reference angles and quadrant signs.

Reciprocals

$\sec=1/\cos$, $\csc=1/\sin$, $\cot=\cos/\sin$; exclude zero denominators.

Example 1

Convert 225° to radians and find its sine and cosine.

Solution

1. $225\pi/180 = 5\pi/4$; reference angle $\pi/4$, quadrant III.

2. $\sin(5\pi/4) = \cos(5\pi/4) = -\sqrt{2}/2$.

Radians and the Unit Circle

Example 2

For $r = 6$ cm and $\theta = 2\pi/3$, find arc length and sector area.

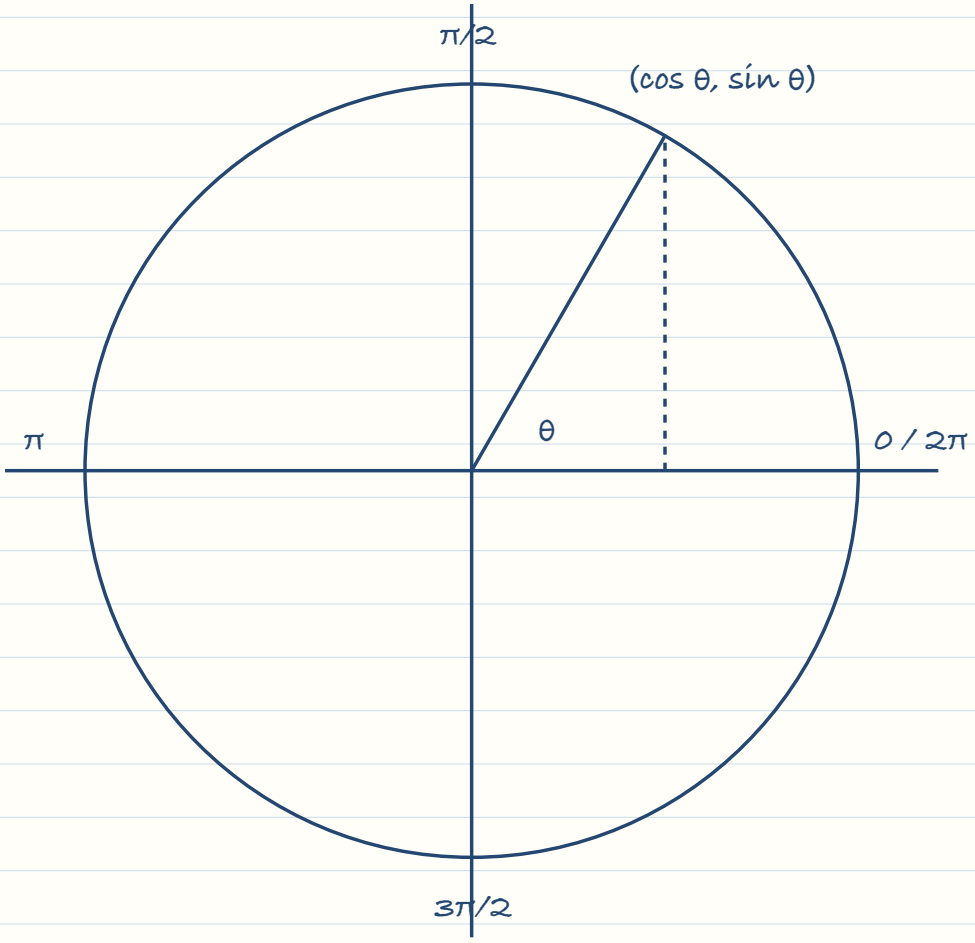
Solution

1. $s = 6(2\pi/3) = 4\pi$ cm.

2. $A = (1/2)(36)(2\pi/3) = 12\pi$ cm^2.

Radians and the Unit Circle

Visual notes | unit circle



The radius is 1. The horizontal coordinate is cosine and the vertical coordinate is sine. The drawn angle is π/3 (60°), giving the exact point (1/2, √3/2).

Signs follow the coordinates: upper half means positive sine; right half means positive cosine. Tangent is sin θ/cos θ and is undefined on the vertical axis.

Radians and the Unit Circle

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Convert $150°$ to radians and $7\pi/6$ to degrees.

2. Find $\sin(5\pi/6)$, $\cos(5\pi/6)$, $\tan(5\pi/6)$.

3. Find the arc length for $r = 8$ m and $\theta = \pi/4$.

4. State where $\tan\theta$ is undefined.

Radians and the Unit Circle

Practice answers | check after trying

1. $5\pi/6$ and $210°$.

2. $1/2$, $-\sqrt{3}/2$ and $-1/\sqrt{3}$, respectively.

3. $s = 8(\pi/4) = 2\pi$ m.

4. $\theta = \pi/2 + k\pi$, k any integer, where $\cos\theta = 0$.

Trigonometric Graphs and Models

Concept

Sine and cosine model repeating changes. Their graphs have a fixed midline and amplitude.

Model

$y = a\sin(b(x-h)) + k$, or cosine.

Features

Amplitude $|a|$; period $2\pi/|b|$; shift h; midline k.

Range

$[k-|a|, k+|a|]$; plot five points one quarter-period apart.

Tangent

Period $\pi/|b|$; unbounded range, so no amplitude.

Example 1

Describe $y = 3\sin(2(x - \pi/4)) - 1$.

Solution

1. Amplitude 3; period π; shift right $\pi/4$; midline -1.
2. Range $[-4,2]$. Quarter-period $= \pi/4$.
3. One-cycle y-values: $-1,2,-1,-4,-1$, starting at $x=\pi/4$.

Trigonometric Graphs and Models

Example 2

A rider moves from 2 m to 18 m in a 40 s rotation, starting at the bottom.

Solution

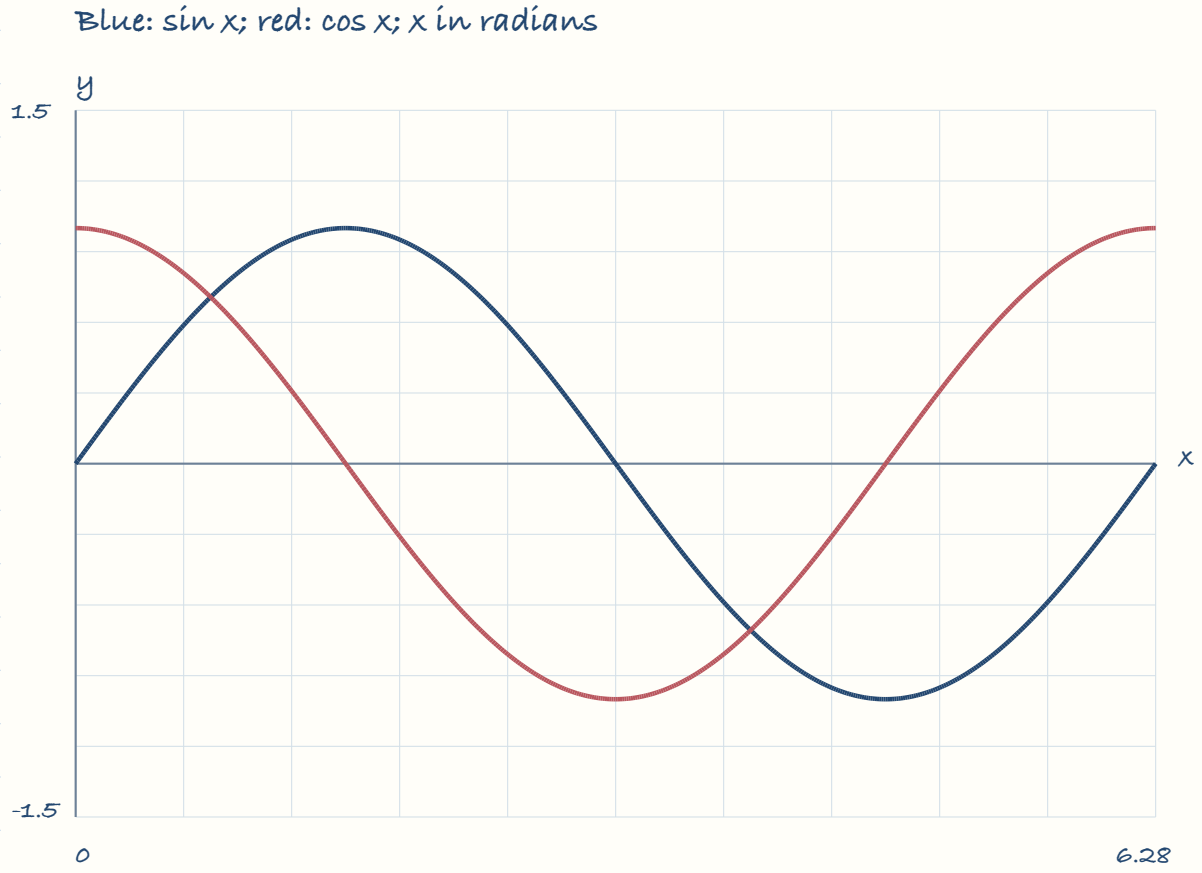
1. Midline = (18+2)/2 = 10; amplitude = (18-2)/2 = 8.

2. $b = 2\pi/40 = \pi/20$.

3. $h(t) = 10 - 8\cos(\pi t/20)$; at $t=20$, $h=18$ m.

Trigonometric Graphs and Models

Visual notes | read the graph



From $x = 0$ to 2π (about 6.283), each graph completes one cycle. Both have amplitude 1 and midline 0.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Trigonometric Graphs and Models

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. State amplitude, period and midline for $y = 4\cos(3x) + 2$.

2. Find range and period of $y = -\sin(x/2)$.

3. Write a cosine model with max 12, min 4, period 6 and a maximum at $t = 0$.

4. Explain why a sinusoid may only approximately model outdoor temperature.

Trigonometric Graphs and Models

Practice answers | check after trying

1. Amplitude 4; period $2\pi/3$; midline $y = 2$.

2. Range $[-1,1]$; period 4π.

3. $y = 8 + 4\cos(\pi t/3)$.

4. Weather, seasonal shifts and unequal heating/cooling can change amplitude, timing and shape, so exact repetition is only an assumption.

Trigonometric Equations

Concept

An inverse trig key gives one value. Use symmetry and periods to find all solutions.

General sine solutions

$x=a+2k\pi$ or $x=\pi-a+2k\pi$, k any integer.

Cosine / tangent

$x=\pm a+2k\pi$ for cosine; $x=a+k\pi$ for tangent.

Multiple arguments

Transform the interval for $bx+c$ before finding angles.

Quadratic trig

Solve in $\sin x$ or $\cos x$ first; reject outputs outside $[-1,1]$.

Example 1

Solve $\cos(2x) = 1/2$ on $0 \le x < 2\pi$.

Solution

1. The argument interval is $0 \le 2x < 4\pi$.
2. $2x = \pi/3, 5\pi/3, 7\pi/3, 11\pi/3$.
3. Divide by 2: $x = \pi/6, 5\pi/6, 7\pi/6, 11\pi/6$.

Trigonometric Equations

Example 2

Solve $2\cos^2 x - \cos x - 1 = 0$ on $0 \le x < 2\pi$.

Solution

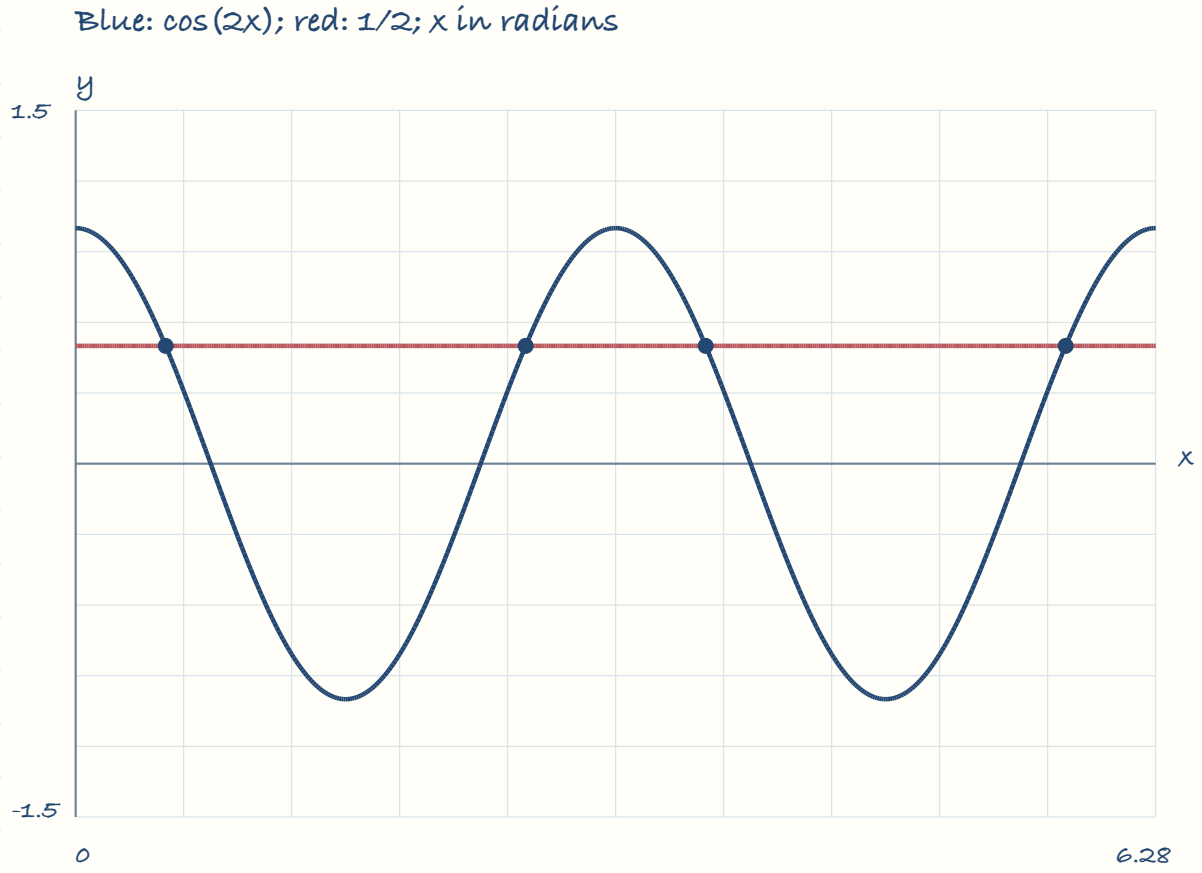
1. Factor: $(2\cos x + 1)(\cos x - 1) = 0$.

2. $\cos x = -1/2$ or 1.

3. $x = 2\pi/3, 4\pi/3, 0$. Do not include the excluded endpoint 2π.

Trigonometric Equations

Visual notes | read the graph



There are four intersections on $[0,2\pi)$. The doubled argument completes two cycles in this interval.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Trigonometric Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $\sin x = -\sqrt{2}/2$ on $[0,2\pi)$.

2. Solve $2\sin^2 x - 1 = 0$ on $[0,2\pi)$.

3. Solve $\tan(2x) = 0$ for all real x.

4. Does $\cos x = 1.2$ have a real solution?

5. Solve $\cos x = 0$ on $[0,2\pi]$.

Trigonometric Equations

Practice answers | check after trying

1. $5\pi/4$ and $7\pi/4$.

2. $\sin x = \pm\sqrt{2}/2$, so $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$.

3. $2x = k\pi$ gives $x = k\pi/2$, k any integer.

4. No: cosine is between -1 and 1.

5. $\pi/2$ and $3\pi/2$; neither endpoint solves the equation.

Trigonometric Identities

Concept

An identity is true wherever both sides are defined. Prove it by transforming one side.

Pythagorean

$\sin^2x+\cos^2x=1;\ 1+\tan^2x=\sec^2x;\ 1+\cot^2x=\csc^2x.$

Sum / difference

$\sin(A\pm B)=\sin A\cos B\pm\cos A\sin B.$

Cosine sum / difference

$\cos(A\pm B)=\cos A\cos B\mp\sin A\sin B.$

Double angle

$\sin2x=2\sin x\cos x;\ \cos2x=1-2\sin^2x=2\cos^2x-1.$

Tangent formula

$\tan(A\pm B)=(\tan A\pm\tan B)/(1\mp\tan A\tan B)$, where defined.

Example 1

Find $\sin75°$ exactly.

Solution

1. $75°=45°+30°$; apply the sine sum identity.
2. $\sin75°=(\sqrt2/2)(\sqrt3/2)+(\sqrt2/2)(1/2)$.
3. Result: $(\sqrt6+\sqrt2)/4$.

Trigonometric Identities

Example 2

Prove $(1 - \cos^2 x)/\sin x = \sin x$ on its original domain.

Solution

1. Restriction: $\sin x \neq 0$.

2. Use $1 - \cos^2 x = \sin^2 x$.

3. Left side $= \sin^2 x/\sin x = \sin x$, as required.

Trigonometric Identities

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find $\cos 75°$ exactly.

2. Simplify $\sec^2 x - \tan^2 x$ on its domain.

3. Prove $(1 + \tan^2 x)\cos^2 x = 1$ on the original domain.

4. If $\cos x = 5/13$ and x is in quadrant IV, find $\sin 2x$.

5. Explain why $\sin(A + B)$ is not $\sin A + \sin B$.

Trigonometric Identities

Practice answers | check after trying

1. $\cos(45° + 30°) = (\sqrt{6} - \sqrt{2})/4$.

2. 1, with $\cos x \neq 0$.

3. Replace $1 + \tan^2 x$ by $\sec^2 x = 1/\cos^2 x$, then multiply; $\cos x \neq 0$ is required.

4. $\sin x = -12/13$, so $\sin 2x = 2(-12/13)(5/13) = -120/169$.

5. At $A = B = 30°$, the left is $\sin 60° = \sqrt{3}/2$, while the right is 1. A single counterexample disproves the proposed identity.